







## **When Concurrency Matters: Behaviour-Oriented Concurrency**

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The state of BoC today



$$\text{STEP} \frac{E, h \hookrightarrow E', h'}{R \uplus (\bar{\kappa}, E), P, h \rightsquigarrow R \uplus (\bar{\kappa}, E'), P, h'}$$

$$\text{SPAWN} \frac{E \hookrightarrow_{\text{when } (\bar{\kappa}') \{E''\}} E'}{R \uplus (\bar{\kappa}, E), P, h \rightsquigarrow R \uplus (\bar{\kappa}, E'), P : (\bar{\kappa}', E''), h}$$

$$\text{RUN} \frac{(\cup_{(\bar{\kappa}', \_) \in (P' \cup R)} \bar{\kappa}') \cap \bar{\kappa} = \emptyset}{R, P' : (\bar{\kappa}, E) : P'', h \rightsquigarrow R \uplus (\bar{\kappa}, E), P' : P'', h}$$

$$\text{END} \frac{\text{finished}(E)}{R \uplus (\bar{\kappa}, E), P, h \rightsquigarrow R, P, h}$$



# Behavior-Oriented Concurrency in Python

Lock-less, Deadlock-free, Ownership-based



<https://microsoft.github.io/bocpy/>











OPSLA 2023

operational semantics



# The state of BoC today

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## Operational Semantics

$$\begin{array}{l} \text{STEP} \frac{E, h \hookrightarrow E', h'}{R \uplus (\bar{\kappa}, E), P, h \rightsquigarrow R \uplus (\bar{\kappa}, E'), P, h'} \quad \text{SPAWN} \frac{E \hookrightarrow_{\text{when } (\bar{\kappa}') \{E''\}} E'}{R \uplus (\bar{\kappa}, E), P, h \rightsquigarrow R \uplus (\bar{\kappa}, E'), P : (\bar{\kappa}', E''), h} \\ \\ \text{RUN} \frac{(\bigcup_{(\bar{\kappa}', \_) \in (P' \cup R)} \bar{\kappa}') \cap \bar{\kappa} = \emptyset}{R, P' : (\bar{\kappa}, E) : P'', h \rightsquigarrow R \uplus (\bar{\kappa}, E), P' : P'', h} \quad \text{END} \frac{\text{finished}(E)}{R \uplus (\bar{\kappa}, E), P, h \rightsquigarrow R, P, h} \end{array}$$

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But, ...

# We specified *how*, and not *what*

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**How a program runs?**

Operational semantics and implementations